53. Galois Theory

1. We have $\{K : \mathbb{Q}\} = [K : \mathbb{Q}] = 8$.

2. We have $|G(K/\mathbb{Q})| = [K : \mathbb{Q}] = 8$ because K is a normal extension of $\mathbb{Q}$.

3. We have $|\lambda(\mathbb{Q})| = |G(K/\mathbb{Q})| = 8$.

4. We have $|\lambda(\mathbb{Q}(\sqrt{2}, \sqrt{3}))| = [K : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 2$.

5. We have $|\lambda(\mathbb{Q}(\sqrt{6}))| = [K : \mathbb{Q}(\sqrt{6})] = 4$.

6. We have $|\lambda(\mathbb{Q}(\sqrt{30}))| = [K : \mathbb{Q}(\sqrt{30})] = 4$.

7. We have $|\lambda(\mathbb{Q}(\sqrt{2} + \sqrt{6}))| = [K : \mathbb{Q}(\sqrt{2} + \sqrt{6})] = 2$, because $\deg(\sqrt{2} + \sqrt{6}, \mathbb{Q}) = 4$.

8. We have $|\lambda(K)| = [K : K] = 1$.

9. Now $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$ so the splitting field of $x^4 + 1$ over $\mathbb{Q}$ is the same as the splitting field of $x^2 + 1$ over $\mathbb{Q}$. This splitting field is $\mathbb{Q}(i)$. It is of degree 2 over $\mathbb{Q}$, and its Galois group is cyclic of order 2 with generator σ where $\sigma(i) = -i$.

10. Because $729 = 9^3$, Theorem 53.7 shows that the Galois group of GF(729) over GF(9) is cyclic of order 3, generated by σ_9 where $\sigma_9(\alpha) = \alpha^9$ for $\alpha \in$ GF(729).

11. See the answer in the text. The answer to Exercise 23 of Section 50 in this manual explains why the group is isomorphic to S_3. We might explain the statement in the text answer that the notation was chosen to reflect the notation in Example 8.7, where S_3 consisted of permutations of $\{1, 2, 3\}$. Here, we are permuting $\{\alpha_1, \alpha_2, \alpha_3\}$, and we defined our permutations so they have the effect on the subscripts 1, 2, and 3 that S_3 has on the numbers 1, 2, and 3. It is worth indicating how this can be verified. Note that $\zeta = \frac{-1+i\sqrt{3}}{2} = -\frac{1}{2} + i\frac{\sqrt{3}}{2} = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$ is a cube root of unity, and $\frac{-1-i\sqrt{3}}{2} = \zeta^2$. Thus our three zeros of $x^3 - 2$ can be written as

$$\alpha_1, \quad \alpha_2 = \alpha_1\zeta, \quad \text{and} \quad \alpha_3 = \alpha_1\zeta^2.$$

According to the definitions of the six automorphism in the text, we see each ρ_i maps ζ into ζ, but each μ_i maps ζ into ζ^2.

We illustrate with two computations how our choice of notation mirrors the effect of our notation for S_3 in Example 8.7. In that example, ρ_1 maps 1 to 2, 2 to 3, and 3 to 1. Here we have $\rho_1(\alpha_1) = \alpha_2$ by definition. Computing we find

$$\rho_1(\alpha_2) = \rho_1(\alpha_1\zeta) = \rho_1(\alpha_1)\rho_1(\zeta) = \alpha_2\zeta = (\alpha_1\zeta)\zeta = \alpha_3,$$

and

$$\rho_1(\alpha_3) = \rho_1(\alpha_1\zeta^2) = \rho_1(\alpha_1)(\rho_1(\zeta))^2 = \alpha_2\zeta^2 = (\alpha_1\zeta)\zeta^2 = \alpha_1$$

because $\zeta^3 = 1$. For our second illustration, we use μ_2, which we expect to leave subscript 2 alone and swap subscripts 1 and 3. Remember that $\mu_2(\zeta) = \zeta^2$, and $\mu_2(\alpha_1) = \alpha_3$. Computing,

$$\mu_2(\alpha_2) = \mu_2(\alpha_1\zeta) = \mu_2(\alpha_1)\mu_2(\zeta) = \alpha_3\zeta^2 = (\alpha_1\zeta^2)\zeta^2 = \alpha_1\zeta = \alpha_2$$

and

$$\mu_2(\alpha_3) = \mu_2(\alpha_1\zeta^2) = \mu_2(\alpha_1)\mu_2(\zeta^2) = \alpha_3\zeta^4 = (\alpha_1\zeta^2)\zeta^4 = \alpha_1$$

because $\zeta^6 = (\zeta^3)^2 = 1^2 = 1$.

12. Now $x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$ so the spitting field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Its Galois group over $\mathbb{Q}$ is isomorphic to the Klein 4-group $\mathbb{Z}_2 \times \mathbb{Z}_2$. See Example 53.3 for a description of the action of each element on $\sqrt{2}$ and on $\sqrt{3}$.

13. Now $x^3 - 1 = (x - 1)(x^2 + x + 1)$. Because a primitive cube root of unity is $\frac{-1+i\sqrt{3}}{2}$, we see that its splitting field over $\mathbb{Q}$ is $\mathbb{Q}(i\sqrt{3})$. The Galois group is cyclic of order 2 and is generated by σ where $\sigma(i\sqrt{3}) = -i\sqrt{3}$.

14. Let $F = \mathbb{Q}, K_1 = \mathbb{Q}(\sqrt{2})$ and $K_2 = \mathbb{Q}(i)$. The fields are not isomorphic because the additive inverse of unity is a square in K_2 but is not a square in K_1. However, the Galois groups over $\mathbb{Q}$ are isomorphic, for they are both cyclic of order 2.

15. F F T T T F F T F T

16. Because $F(K/E) \leq G(K/F)$ and $G(K/F)$ is abelian, we see that $G(K/E)$ is abelian, for a subgroup of an abelian group is abelian. Because $G(E/F) \simeq G(K/F)/G(K/E)$ and $G(K/F)$ is abelian, we see that $G(E/F)$ is abelian, for a factor group of an abelian group,where multiplication is done by choosing representatives, must again be abelian.

17. To show that $N_{K/F}(\alpha) \in F$, we need only show that it is left fixed by each $\tau \in G(K/F)$. From the given formla and the fact that τ is an automorphism, we have

$$\tau(N_{K/F}(\alpha)) = \prod_{\sigma \in G(K/F)} (\tau\sigma)(\alpha).$$

But as σ runs through the elements of $G(K/F), \tau\sigma$ again runs through all elements, because $G(K/F)$ is a group. Thus only the order of the factors in the product is changed, and because multiplication in K is commutative, the product is unchanged. Thus $\tau(N_{K/F}(\alpha)) = N_{K/F}(\alpha)$ for all $\tau \in G(K/F)$, so $N_{K/F}(\alpha) \in F$. Precisely the same argument shows that $Tr_{K/F}(\alpha) \in F$, only this time it is the order of the summands in the sum that gets changed when computing $\tau(Tr_{K/F}(\alpha))$.

18. **a.** $N_{K/\mathbb{Q}}(\sqrt{2}) = \sqrt{2}\sqrt{2}(-\sqrt{2})(-\sqrt{2}) = 4$, because two of the elements of the Galois group leave $\sqrt{2}$ fixed, and two carry it into $-\sqrt{2}$. The computations shown for Part(**b**) - Part(**h**) are based similarly on action of the Galois group on $\sqrt{2}$ and $\sqrt{3}$.

b. $N_{K/\mathbb{Q}}(\sqrt{2} + \sqrt{3}) = (\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3})(-\sqrt{2} + \sqrt{3})(-\sqrt{2} - \sqrt{3}) = (-1)^2 = 1.$

c. $N_{K/\mathbb{Q}}(\sqrt{6}) = (\sqrt{6})(-\sqrt{6})(-\sqrt{6})(\sqrt{6}) = 36$

d. $N_{K/\mathbb{Q}}(2) = (2)(2)(2)(2) = 16$

e. $Tr_{K/\mathbb{Q}}(\sqrt{2}) = \sqrt{2} + \sqrt{2} + (-\sqrt{2}) + (-\sqrt{2}) = 0$

f. $Tr_{K/\mathbb{Q}}(\sqrt{2} + \sqrt{3}) = (\sqrt{2} + \sqrt{3}) + (-\sqrt{2} + \sqrt{3}) + (\sqrt{2} - \sqrt{3}) + (-\sqrt{2} - \sqrt{3}) = 0.$

g. $Tr_{K/\mathbb{Q}}(\sqrt{6}) = \sqrt{6} + (-\sqrt{6}) + (-\sqrt{6}) + \sqrt{6} = 0$

h. $Tr_{K/\mathbb{Q}}(2) = 2 + 2 + 2 + 2 = 8$

19. Let $f(x) = \text{irr}(\alpha, F)$. Because $K = F(\alpha)$ is normal over F, it is a splitting field of $f(x)$. Let the factorization of $f(x)$ in $K[x]$ be

$$f(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n) \tag{1}$$

where $\alpha = \alpha_1$. Now $|G(K/F)| = n$, and α is carried onto each α_i for $i = 1, 2, \cdots, n$ by precisely one element of $G(K/G)$. Thus $N_{K/F}(\alpha) = \alpha_1\alpha_2 \cdots \alpha_n$. If we multiply the linear factors in (1)

together, we see that the constant term a_0 is $(-1)^n\alpha_1\alpha_2\cdots\alpha_n = (-1)^n N_{K/F}(\alpha)$. Similarly, we see that $Tr_{K/F}(\alpha) = \alpha_1 + \alpha_2 + \cdots + \alpha_n$. If we pick up the coefficient a_{n-1} of x^{n-1} in $f(x)$ by multiplying the linear factors in (1), we find that $a_{n-1} = -\alpha_1 - \alpha_1 - \cdots - \alpha_n = -Tr_{K/F}(\alpha)$.

20. Let $\alpha_1, \alpha_2, \cdots, \alpha_r$ be the distinct zeros of $f(x)$ in $\overline{F}$ and form the splitting field $K = F(\alpha_1, \alpha_2, \cdots, \alpha_r)$ of $f(x)$ in $\overline{F}$. Note that $r \leq n$ because $f(x)$ has at most n distinct zeros. Because all irreducible factors of $f(x)$ are separable, we see that K is normal over F. Now each $\sigma \in G(K/F)$ provides a permutation of $S = \{\alpha_1, \alpha_2, \cdots, \alpha_r\}$ and distinct elements of $G(K/F)$ correspond to distinct permutations of S because an automorphism of K leaving F fixed is uniquely determined by its values on the elements of S. Because permutation multiplication and multiplication in $G(K/F)$ are both function composition, we see that $G(K/F)$ is isomorphic to a subgroup of the group of all permutations of S, which is isomorphic to a subgroup of S_r. By the Theorem of Lagrange, it follows that $|G(K/F)|$ divides $r!$ which in turn divides $n!$ because $r \leq n$.

21. Let K be the splitting field of $f(x)$ over F. Because each $\sigma \in G(K/F)$ carries a zero of $f(x)$ into a zero of $f(x)$ and is a one-to-one map, it induces a permutation of the set S of distinct zeros in $\overline{F}$ of $f(x)$. The action of $\sigma \in G(K/G)$ on all elements of K is completely determined by its action on the elements of the set S. Because permutation multiplication and multiplication in $G(K/F)$ are both function composition, we see that $G(K/F)$ is isomorphic in a natural way to a subgroup of the group of all permutations of S.

22. **a.** Exercise 17 of Section 51 shows that $x^n - 1$ has no zeros of multiplicity greater than 1 as long as $n \cdot 1$ is not equal to zero in F. Thus the splitting field of $x^n - 1$ over F is a normal extension. If ζ is a primitive nth root of unity, then $1, \zeta, \zeta^2, \cdots, \zeta^{n-1}$ are distinct elements, and are all zeros of $x^n - 1$. Thus the splitting field of $x^n - 1$ over F is $F(\zeta)$.

b. The action of $\sigma \in G(F(\zeta)/F)$ is completely determined by $\sigma(\zeta)$ which must be one of the conjugates ζ^s of ζ over F. Let $\sigma, \tau \in G(F(\zeta)/F)$, and suppose $\sigma(\zeta) = \zeta^s$ and $\tau(\zeta) = \zeta^t$. Then

$$\begin{aligned}(\sigma\tau)(\zeta) &= \sigma(\tau(\zeta)) = \sigma(\zeta^t) = [\sigma(\zeta)]^t = (\zeta^s)^t = \zeta^{st} = \zeta^{ts} \\ &= (\zeta^t)^s = [\tau(\zeta)]^s = \tau(\zeta^s) = \tau(\sigma(\zeta)) = (\tau\sigma)(\zeta)\end{aligned}$$

so $\sigma\tau = \tau\sigma$ and $G(F(\zeta)/F)$ is abelian.

23. **a.** Because K is cyclic over F, we know that $G(K/F)$ is a cyclic group. Now $G(K/E)$ is a subgroup of $G(K/F)$, and is thus cyclic as a subgroup of a cyclic group. Therefore K is cyclic over E. Because E is a normal extension of F, we know that $G(E/F) \simeq G(K/F)/G(K/E)$ so $G(E/F)$ is isomorphic to a factor group of a cyclic group, and is thus cyclic. (A factor group of a cyclic group A is generated by a coset containing a generator of A.) Therefore E is cyclic over F.

b. By Galois theory, we know that there is a one-to-one correspondence between subgroups H of $G(K/F)$ and fields $E = K_H$ such that $F \leq E \leq K$. Because $G(K/F)$ is cyclic, it contains precisely one subgroup of each order d that divides $|G(K/F)| = [K : F]$. Such a subgroup corresponds to a field E where $F \leq E \leq K$ and $[K : E] = d$, so that $[E : F] = m = n/d$. Now as d runs through all divisors of n, the quotients $m = n/d$ also run through all divisors of n, so we are done.

24. **a.** For $\tau \in G(K/F)$, we have a natural extension of τ to an automorphism $\overline{\tau}$ of $K[x]$ where $\overline{\tau}(\alpha_0 + \alpha_1 x + \cdots + \alpha_n x^n) = \tau(\alpha_0) + \tau(\alpha_1)x + \cdots + \tau(\alpha_n)x^n$. Clearly the polynomials left fixed by $\overline{\tau}$ for *all* $\tau \in G(K/F)$ are precisely those in $F(x)$. For $f(x) = \prod_{\sigma\in G(K/F)}(x - \sigma(\alpha))$, we have

$$\overline{\tau}(f(x)) = \prod_{\sigma\in G(K/F)} (x - (\tau\sigma)(\alpha)).$$